ZEERBOX · STEM CHECK

ZeerBox STEM check

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DOCUMENT CONTROL

Document ID	ZBX-STM-001	Version	0.1
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/zeerbox		
Source	docs/01b-stem-check.md @ 44ce24e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.9.1 upgrade of a TRL 3 repo): 37 option A claims screened against independent estimates; two detail fixes under the standing instruction (ZBX-DEC-001 D1, D5); options B and C screened at first order; independent review recorded and acted on

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Verdict: the working principle of the existing design (option A, forced air through a wetted cellulose pad into an insulated store, with a controller that reads the weather) sits within known physics, chemistry and biology, and so do the principles of options B and C. Of the 37 claims for option A, 30 are within known science and 7 are plausible. Produce transpiration (S26) is closed by the calculation's scenario range; the other six need supplier data, field data or a decision by Amish: the pad's effectiveness and pressure drop (S3, S5), the shelf-life gain (S25), the battery's charging temperature (S31, ZBX-DEC-001 D6), the roof-space temperature (S35) and ventilation through a long sunless spell (S36, D3). Two details failed and were fixed under Amish's standing instruction of 2026-10-23: the 1.5 mm² load cable was protected only by the 15 A battery fuse (S16, D1), and, found by the independent review, the 4 L sump bleed let the pad water reach about 9 cycles of concentration (S37, D5). No row fails. Because six rows can only be closed by data or a decision, `stem_check` is **open**; under STANDARDS section 20 the repo keeps TRL 3 while they are open. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports the calculation note's script `docs/04-calcs/sizing.py` (ZBX-CAL-001 v0.4) and writes `docs/04-calcs/stem_results.csv`.

The numbers checked are those of ZBX-CAL-001; each check uses a method independent of the one CAL used where one exists (an empirical wet-bulb fit, a Magnus moisture balance, an energy balance on the airflow), and otherwise a limit the claim must sit inside (the wet bulb, the free-air flow of the fans, the electrical input, the Shockley-Queisser limit). Several of option A's headline rows (S6 to S9) rest on the pad effectiveness of S3: the 8 K of PR-002 holds down to about 68 % effectiveness.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv` by `stem_check.py`). S rows are option A, B rows option B and C rows option C (ZBX-OPT-001).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Outside wet-bulb temperature at the 35 °C, 30 % RH design point is 21.5 °C	Physics	Thermodynamics	Psychrometrics; independent check with the Stull (2011) wet-bulb fit	claimed 21.52 °C, estimate 22.1 °C (ratio 0.97)	Within known science	ZBX-CAL-001 s4
S2	Air leaving the pad (23.1 °C) stays above the wet-bulb temperature, the limit of direct evaporative cooling	Physics	Thermodynamics	Adiabatic saturation: direct evaporative cooling cannot go below the wet bulb	23.11 °C >= 21.52 °C	Within known science	ZBX-CAL-001 s4
S3	A 150 mm cellulose pad reaches about 88 % saturation effectiveness at 0.52 m/s face velocity	Physics	Heat transfer	NTU model calibrated to 75 % at 0.6 m/s for 100 mm (CAL assumption); maker data for deeper pads show 90 % or more (CELdek)	Within the range makers publish; the chosen pad's own chart is not yet in hand (ZBX-DEC-001 to confirm 2). The drop is set by it: PR-002 (8 K) holds down to about 68 % effectiveness (independent review)	Plausible, to confirm at TRL 3	Open
S4	Two 400 m³/h fans deliver about 563 m³/h through the pad, shutters and guards	Physics	Fluid mechanics	A bound only: fan delivery cannot exceed the free-air flow of the two fans in parallel (800 m³/h); the operating point itself rests on S5	563 m³/h <= 800 m³/h	Within known science	ZBX-CAL-001 s3
S5	The fan curve and the wet pad's pressure drop give the 18 Pa operating point	Physics	Fluid mechanics	Straight-line fan curve to a 50 to 70 Pa shut-off; wet 100 mm pad 15 Pa at 1.0 m/s	Assumed curves; supplier data for the chosen fan and pad needed (ZBX-	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				(CAL assumptions)	DEC-001 to confirm 2). At only about 18 Pa, a 3 m/s wind on the pad wall (about 5 Pa) moves the flow by 15 to 20 %, and in hold mode (one fan at 30 %, about 5 Pa shut-off) it can stall or reverse the flow (independent review)		
S6	Store air warms about 2.35 K from supply to exhaust while it absorbs the 443 W heat load	Physics	Thermodynamics	Steady-flow energy balance, rise = Q / (rho V cp), rho 1.15 kg/m3	claimed 2.35 K, estimate 2.45 K (ratio 0.96)	Within known science	ZBX-CAL-001 s4
S7	Mean store air is 10.7 K below outside air at the design point	Physics	Thermodynamics	The store cannot be cooler than the supply air, so the drop is below effectiveness x wet-bulb depression (tighter bound, independent review)	10.71 K <= 11.89 K	Within known science	ZBX-CAL-001 s4; rests on S3
S8	The pad evaporates about 32 L of water on the design day (10 h of cooling)	Physics	Thermodynamics	Latent heat taken up equals the sensible heat given up by the air, h_fg 2.43 MJ/kg	claimed 31.9 L, estimate 31.9 L (ratio 1.00)	Within known science	ZBX-CAL-001 s8
S9	Mean store RH is about 81 % while cooling at the design point	Physics	Thermodynamics	Moisture balance with the Magnus saturation pressure	claimed 81.5 %, estimate 81.6 % (ratio 1.00)	Within known science	ZBX-CAL-001 s5; rests on S3 (about 1 point of RH per point of effectiveness)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S10	One 150 W panel covers the 372 Wh design day (10 h cooling, 2 h evening ventilation)	Engineering and technology	Power and link budgets	Energy balance; at 40 % losses (PWM, dust, harmattan haze, cable) the panel gives 450 Wh, still 21 % over (independent review)	in 525 Wh/ day, out 372 Wh/day	Within known science	ZBX-CAL-001 s7
S11	A 150 W panel of 1,480 x 670 mm	Physics	Optics and radiation	Module efficiency at 1,000 W/m2 below the Shockley-Queisser limit for one junction	0.151 <= 0.337	Within known science	BOM line 10
S12	The 12.8 V, 12 Ah battery carries the 62 Wh used after sunset	Chemistry	Electrochemistry and batteries	Energy balance	in 122.9 Wh, out 62 Wh	Within known science	ZBX-CAL-001 s7
S13	A 20 A PWM charge controller suits the 150 W panel	Engineering and technology	Standards practice	Controller rating at least 1.25 x the panel's short-circuit current (9 A); it must also have a LiFePO4 charge profile (about 14.2 to 14.4 V), no lead-acid temperature compensation and a low-voltage disconnect (BOM line 11)	20 A >= 11.25 A	Within known science	ZBX-CAL-001 s7
S14	Two 12 W fans move 563 m³/h at about 18 Pa	Physics	Fluid mechanics	Air power Q x dp cannot exceed the 24 W of electrical input (implied fan efficiency about 12 %, usual for small axial fans)	2.78 W <= 24 W	Within known science	ZBX-CAL-001 s3
S15		Physics	Fluid mechanics		1.18 W <= 6 W		

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	A 6 W pump lifts 3.6 L/min to the header 2 m up			Hydraulic power rho g H Q cannot exceed the electrical input		Within known science	ZBX-CAL-001 s8
S16	The 1.5 mm² load cable in the 45 °C roof space is protected by its fuse	Physics	Electromagnetism and circuits	Fuse rating I_n <= derated cable capacity I_z (IEC 60364-5-52 tables B.52.4 and B.52.14, to confirm against the edition used); 5 A load fuse at the source end of the run after the fix; both fuses DC rated with a breaking capacity above the battery's short-circuit current (D5)	5 A <= 13.8 A	Within known science	ZBX-DEC-001 D1 (fix under the standing instruction); ZBX-CAL-001 s9
S17	Each roof post footing holds down its share of the uplift at a 30 m/s gust	Physics	Mechanics and statics	Footing weight against net uplift per post, with a factor of at least 1.5, at a 30 m/s gust and a mean net Cp of 1.5; the factor falls below 1.5 above about 32 m/s, and edge zones of a canopy reach Cp about 2 (independent review; ZBX-DEC-001 D7)	1.81 >= 1.5	Within known science	ZBX-CAL-001 s10
S18	The 50 x 150 mm purlins carry the uplift on a 0.7 m strip over 3.7 m	Physics	Strength of materials	Simply supported beam, sigma = M / Z; sawn softwood about 7 MPa allowed (grade to confirm); at an edge-zone Cp of 2.0 the stress	5.17 MPa <= 7 MPa	Within known science	ZBX-CAL-001 s10

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				reaches about 6.9 MPa (D7)			
S19	The 75 x 150 mm roof beams span 2.8 m between posts	Physics	Strength of materials	Simply supported beam under the uplift less the sheet weight	5.28 MPa <= 7 MPa	Within known science	ZBX-CAL-001 s10
S20	Each purlin end needs a tie of 2 kN rated uplift	Engineering and technology	Standards practice	Tie load from the purlin's uplift, half at each end; edge purlins carry more and the nails must hold the rating in the timber (D7)	1.08 kN <= 2 kN	Within known science	ZBX-CAL-001 s10; ZBX-DEC-001 to confirm 4
S21	Shelf rails carry four 22 kg crates over 1.05 m	Physics	Strength of materials	Simply supported 50 x 75 mm rail, sigma = M / Z	0.58 MPa <= 7 MPa	Within known science	ZBX-CAL-001 s2
S22	The 7 t of brick bears on a 450 mm strip footing	Physics	Mechanics and statics	Bearing pressure against about 100 kPa allowed on firm soil; not valid on expansive clay or collapsible sand, so the site soil is to check	15.7 kPa <= 100 kPa	Within known science	ZBX-CAL-001 s10
S23	Dry loose rice husk conducts about 0.06 W/(m·K)	Physics	Heat transfer	Measured 0.065 to 0.075 W/(m·K) for rice husk insulators (Cigarruista Solis et al. 2024)	claimed 0.06 W/(m·K), estimate 0.07 W/(m·K) (ratio 0.86)	Within known science	ZBX-CAL-001 s4.1 (at 0.075 the wall gain rises by about 14 W, about 0.1 K in the store; the review recommends 0.07 central and 0.06 to 0.09 in the scenarios at the next CAL revision)
S24	Mixed produce near 25 °C gives off about	Biology	Plant and animal biology	Respiration classes at 5 °C (Kader 2002),	claimed 0.15 W/kg, estimate	Within known science	ZBX-CAL-001 s4 (at 0.22 W/kg the store is

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	0.15 W/kg of respiration heat			about 5 x at 25 °C; 10.7 J per mg CO2 from glucose oxidation	0.222 W/kg (ratio 0.68)		about 0.2 K warmer, inside the unfavorable case at 0.25 W/kg)
S25	Tomato shelf life rises at least 1.5 times over ambient storage in the dry season	Biology	Plant and animal biology	Zero energy cool chambers raised tomato shelf life from 6 to 11 days (e-planet 18(2)); D-Lab Mali reported 10 days against 2 to 4 for peppers and eggplant	Evidence comes from cooler, more humid passive chambers; a paired storage trial with the partner is needed (TRL 4)	Plausible, to confirm at TRL 3	Open
S26	Produce loses 0.5 to 2 % of its mass a day as water vapor, adding about 0.3 g/kg to the airstream	Biology	Plant and animal biology	Transpiration allowance bracketing leafy and fruit crops; it moves store RH by under 2 percentage points	Bracketed by the favorable and unfavorable cases	Plausible, to confirm at TRL 3	ZBX-CAL-001 s5 (scenarios)
S27	A covered, shaded sump drained weekly, a daily bleed and a pad dried out at the end of every cooling day keep Legionella growth low	Biology	Microbiology	WHO, Legionella and the prevention of legionellosis (2007): stagnant warm water (25 to 45 °C), sediment and biofilm favour growth; drain, clean and keep systems dry. Exposure route: the pad's drift is carried through the room past people entering	Controls follow the guidance: daily dry-out (pump off 60 min before the fans stop) and the bleed added (D5); fans run 10 min before entry; disinfection of the sump proposed (D8); sump water never touches produce	Within known science	ZBX-PRC-001 Safety; ZBX-DEC-001 D5
S28	With at least one air change an hour the store's CO2	Biology	Human physiology and biomechanics	Steady-state CO2 rise = produce CO2 / ventilation; 480	2297 ppm <= 5000 ppm	Within known science	ZBX-CAL-001 s6 (hold mode about 8 ACH)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	stays below a 5,000 ppm workplace limit			kg at about 50 mg CO2/(kg h)			
S29	Fired brick in the pad wall stays strong when wet	Chemistry	Materials and polymer chemistry	Fired clay is a ceramic and does not soften in water, unlike unfired earth	Material review	Within known science	ZBX-DEC-001 (fired brick)
S30	Aluminium panel frames with stainless bolts on a galvanised roof do not corrode in a dry inland climate	Chemistry	Electrochemistry and batteries	Zinc and aluminium sit close in the galvanic series; small stainless fasteners in a large aluminium area corrode little in dry, unpolluted air	Aluminium is anodic to stainless, so EPDM isolating washers go under every stainless bolt on the aluminium frames by default (D5); inspection each season	Within known science	ZBX-DEC-001 D5; ZBX-BLD-001 section 3.13
S31	A lithium iron phosphate battery with a BMS, kept shaded and charged between 0 and 45 °C, is safe in this duty	Chemistry	Electrochemistry and batteries	LiFePO4 has a higher thermal-runaway onset than cobalt-based cells; makers limit charging to about 0 to 45 °C. In a shaded box on the outside wall at 35 to 40 °C air, charging at 0.69 C self-heats the pack toward that limit, so the BMS may stop charging at midday (independent review)	Safe, because the BMS stops charging; the energy lost on the hottest days is to assess. Moving the power box into the cooled store is proposed to Amish (ZBX-DEC-001 D6)	Plausible, to confirm at TRL 3	Open
S32	Low-cost sensors resolve the 4 K wet-bulb	Engineering and technology	Sensing resolution	Depression error from +-3 % RH and +-0.5 K at 30 °C, 75 % RH, against a quarter	0.52 K <= 1 K	Within known science	ZBX-CAL-001 s6

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	depression threshold			of the threshold; sun on an unshielded sensor would add 2 to 3 K, so the six-plate shield and a field calibration at TRL 4 are part of the claim			
S33	563 m³/h is 331 cfm and about 78 air changes an hour in the 7.2 m³ room	Mathematics	Units and dimensional analysis	1 cfm = 1.699 m3/h	claimed 331 cfm, estimate 331 cfm (ratio 1.00)	Within known science	ZBX-CAL-001 s3
S34	The favorable to unfavorable range of the drop (9.8 to 11.4 K) brackets the inputs varied in the three scenarios (pad effectiveness, wall wetting, respiration, fan shut-off, leaks, sun, roof space, ground, transpiration); husk conductivity, the pad pressure-drop law, wind on the pad and sensor error are not varied (independent review)	Mathematics	Statistics and uncertainty	Three-scenario bracketing of every uncertain input (CAL Table 1)	claimed 10.71 K, estimate 10.6 K (ratio 1.01)	Within known science	ZBX-CAL-001 s4
S35	The shaded roof space stays near 40 °C (38 to 43 °C) on the design day	Physics	Heat transfer	Allowance for a ventilated 450 mm gap under a sunlit sheet roof; ceiling gain 46 W	Assumption; a roof-space temperature log is needed (TRL 4)	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				in the central case			
S36	The battery covers the controller and hold-mode ventilation through cloudy spells	Engineering and technology	Power and link budgets	123 Wh usable; hold mode about 2 W plus the 1 W controller lasts about 40 h with no sun; beyond that the fans stop and the room is sealed	A passive vent for the no-power case is proposed to Amish (ZBX-DEC-001 D3)	Plausible, to confirm at TRL 3	Open
S37	The daily bleed keeps the pad water at no more than 4 cycles of concentration, so hard water does not scale the pad	Chemistry	Inorganic chemistry	Cycles of concentration = (evaporation + bleed) / bleed; cellulose pads scale above about 3 to 5 cycles in typical water (independent review). The TRL 3 design's 4 L bleed gave about 9 cycles; fixed (D5)	4 cycles <= 4 cycles	Within known science	ZBX-DEC-001 D5; ZBX-CAL-001 s8 (bleed 10.6 L, total 43 L/day)
B1	Option B holds the full room about 3 K below outside air at the design point	Physics	Heat transfer	Wall-to-wet-cavity balance with stack ventilation; ZECCs measured 4.4 to 5.0 K monthly mean, 7.25 K best day (e-planet 18(2))	claimed 3 K, estimate 3.16 K (ratio 0.95)	Within known science	ZBX-OPT-001 first-order numbers
B2	Option B uses about 70 L of water a day	Physics	Heat transfer	Evaporation from the wet outer leaf (Lewis relation) plus the cavity's conduction loss	claimed 70 L, estimate 61.2 L (ratio 1.14)	Within known science	ZBX-OPT-001 first-order numbers
B3	A low vent and a roof stack of 0.02 m² each give option B at	Physics	Fluid mechanics	Stack flow Q = Cd A sqrt(2 g h dT / T), Cd 0.6, 1.6 m between openings; the room is cooler	3.4 ACH >= 1 ACH	Within known science	ZBX-OPT-001 first-order numbers

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	least one air change an hour			than outside, so cool air drains out of the low vent and 35 °C air enters at the stack (direction noted by the independent review)			
B4	Option B keeps the room humid (55 to 90 % RH)	Physics	Thermodynamics	A moisture balance with produce transpiration alone and the B3 flow gives about 55 % RH (independent review); ZECCs report high humidity because their inner faces are wet	Depends on how wet the inner leaf runs; to assess	Plausible, to confirm at TRL 3	Open
B5	A damp, slowly ventilated chamber grows mold on walls and produce unless cleaned and aired	Biology	Microbiology	High humidity and little air movement favour mold; D-Lab notes cleaning and airing practice	Practice; no growth figure claimed	Plausible, to confirm at TRL 3	Open
C1	Option C holds the room about 7 K below outside air at the design point, in a light wind	Physics	Thermodynamics	Wet-wall effectiveness 0.75 (Wayua et al. 2012 measured 74 to 87 % and 10.5 K at 14:00) and a heat balance on the ventilator flow	First-order 6.9 K at 250 m³/h; conditional on C2, so it cannot be closed while C2 is open (independent review)	Plausible, to confirm at TRL 3	Open
C2	A 300 mm turbine ventilator draws about 250 m³/h in a light wind	Physics	Fluid mechanics	Wind-driven extraction; no rated airflow found in the maker listing checked (MKT);	To assess with the maker's chart; option C's cooling scales with it	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				calm days give little flow			
C3	Option C uses about 40 L of water a day	Physics	Thermodynamics	Latent heat balance on the ventilator flow plus drip losses of about 20 L	claimed 40 L, estimate 32 L (ratio 1.25)	Within known science	ZBX-OPT-001 first-order numbers
C4	Galvanised mesh lasts in permanently wet charcoal	Chemistry	Electrochemistry and batteries	Wet carbon is an electrical conductor and noble to zinc; contact can speed corrosion of the galvanising	Mesh life to assess; stainless or plastic mesh is the fallback	Plausible, to confirm at TRL 3	Open
C5	Wet charcoal walls stay free of mold and smell	Biology	Microbiology	Permanently wet organic-carbon media can grow mold; charcoal coolers are washed and the charcoal replaced	Practice to confirm with users	Plausible, to confirm at TRL 3	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. A Plausible row with a "Closed by" entry is closed by that calculation; "Open" means it needs data that only a TRL 4 bench or field step can give, or a decision by Amish.

Fixes made under the standing instruction

Item	What failed	Fix (ZBX-DEC-001)	Result
Load-circuit cable protection (S16)	ZBX-CAL-001 v0.3 said the 15 A battery terminal fuse sat below the cable ratings. For the 1.5 mm² load cable it does not: two loaded PVC conductors in conduit carry 17.5 A at 30 °C (IEC 60364-5-52, method B1) and 0.79 of that in a 45 °C roof space, 13.8 A, so a fault on the load circuit could overheat the cable before the fuse blew	D1: 2.5 mm² battery leads (18.9 A derated) behind the 15 A battery fuse, and a separate 5 A fuse on the load circuit at the power box; about USD 1 (estimate) in BOM line 11. D5 (b): both fuses DC rated with a breaking capacity above the battery's short-circuit current	5 A against 13.8 A (S16); kit USD 300, at the value-engineering target; CAL section 9 and PR-009 updated
Sump bleed and scale (S37, from the independent review)	4 L of bleed against 32 L evaporated concentrates the sump water about 9 times; cellulose pads scale above about 3 to 5 cycles, and the site water may be hard	D5 (a): bleed sized for at most 4 cycles, 10.6 L a day, drained before topping up	Total water 43 L a day (was 36 L), PR-008 still met (70 L); the sump still holds a day

Proposed, awaiting Amish

- **S36, no-power ventilation (ZBX-DEC-001 D3).** The battery runs hold-mode ventilation and the controller for about 40 h with no sun. Beyond that the fans stop and, with the door shut, the room is sealed: 480 kg of produce respiring at about 50 mg CO_2/(kg·h) takes the air to about 2.3 % CO_2 in 12 h. The independent review added that in hold mode one fan at 30 % (about 5 Pa) can be stalled by a 3 m/s wind on the pad wall. Recommended: an always-open meshed vent and stack with a turbine ventilator head (the graft from options B and C). It lets some cooled air out on design days, so it is a safety trade-off for Amish.
- **S31, battery charging temperature (D6).** Recommended: move the power box into the store, where the air is about 24 °C.
- **S17, S18, S20, design gust and pressure zones (D7).** Recommended: keep 30 m/s until the site's design gust is known, with a hold point before the roof is built.
- **S27, sump disinfection (D8).** Recommended: a weekly shock dose of chlorine after cleaning.

Fields not relevant

None. All five fields are touched: physics (thermodynamics, heat transfer, fluid mechanics, structures, circuits, optics of the panel), chemistry (battery, materials, corrosion, scale), biology (produce respiration and transpiration, microbiology of the sump, human physiology in a closed room), mathematics (units, scenario bracketing) and engineering (power budget, sensing, standards practice). Within physics, nuclear and ionising radiation and acoustics are not relevant: nothing in the design emits radiation, and the 12 W fans are not a noise hazard at the 2.5 m between the fan box and a person outside.

Concept options

- **Option A, forced-air pad store (the existing design):** within known science; 7 plausible rows, one closed by the scenario range and six needing data or a decision; two detail fixes made (D1, D5).
- **Option B, passive wet-wall chamber:** within known science on cooling, water and ventilation (B1 to B3); a full room with 480 kg of produce and the day's warm field produce is held only about 3 K below outside air at the design point (first-order balance; ZECCs with lighter loads show 4 to 7 K), below the 8 K of PR-002; its stack flow drains cool air out of the low vent and draws 35 °C air in at the top. Its humidity (B4, 55 to 90 % RH depending on how wet the inner leaf runs) and mold in a damp, slowly ventilated room (B5) are plausible.
- **Option C, charcoal-wall wind cooler:** about 7 K in a light wind (C1), conditional on the turbine ventilator's airflow, for which no rated figure was found (C2), so both stay plausible; water use (C3) is within known science; the galvanised mesh in permanently wet charcoal (C4) and mold in the charcoal (C5) are plausible risks to confirm.

No option has a Fails row, so none is dropped.

Independent review

A second agent, run headless with only this document, the calculation note (ZBX-CAL-001), the product requirements (ZBX-PRS-001) and the sizing script in front of it and none of the author's reasoning, tried to refute the headline results and every "Within known science" row (2026-10-24). It recomputed by hand and fetched no sources. What it found and what changed:

Finding	Reviewer's verdict	What changed
The 10.7 K drop is set almost entirely by the unconfirmed pad effectiveness; PR-002 fails below about 70 %	Weak	S3 now states the threshold (about 68 %); S7 and S9 marked as resting on S3; supplier chart still to get (to confirm 2)
The heat balance leaves out the latent heat of produce transpiration (about 135 W leaves as vapour)	Weak, conservative	Recorded as a conservative omission (about 0.35 K); no change to the CAL figures
Wind on the pad wall (about 5 Pa at 3 m/s) is the same order as the 18 Pa operating point; hold mode can stall	Weak	Added to S5 and to D3, which it strengthens
The 4 L bleed gives about 9 cycles of concentration; pads scale above 3 to 5	Weak	**Fixed** (D5 a): 10.6 L bleed, 43 L a day; new row S37
A 30 % solar loss is tight once PWM, dust and harmattan haze are counted	Stands	S10 now shows a 40 % loss case: 450 Wh, still 21 % over the need
The fuses must be DC rated with enough breaking capacity; the load fuse at the source end	Stands, with gaps	**Fixed** (D5 b) in BOM line 11, CAL section 9 and S16
Footing factor falls below 1.5 above about 32 m/s; canopy edge zones reach Cp about 2, purlins about 6.9 MPa	Weak beyond 30 m/s	Noted in S17, S18 and S20; proposed to Amish as D7
S4 and S14 are bounds that cannot fail; S7's bound should be effectiveness x depression	Weak	S4 relabelled as a bound resting on S5; S7 now uses the tighter 11.9 K bound
S13 must also require a LiFePO4 charge profile and a low-voltage disconnect	Weak	Added to S13
Rice husk conductivity central 0.07 rather than 0.06; respiration 0.22 rather than 0.15 W/kg	Weak	Effects quantified in S23 (+14 W, about 0.1 K) and S24 (about 0.2 K, inside the unfavorable case); the next CAL revision should adopt them
Legionella: the exposure route (pad drift through the room) and the pad being wet all day were missing	Weak	**Fixed** (D5 d): daily pad dry-out and 10 min of fans before entry; disinfection proposed (D8)
CO_2 at one air change: 1.80 g/L at 25 °C, not 1.96	Stands, arithmetic	Corrected: about 2,300 ppm at 1 air change an hour, 2.3 % sealed for 12 h
Aluminium is anodic to stainless; "dry inland" is an assumption	Weak	**Fixed** (D5 c): EPDM isolating washers by default
The battery may reach its 45 °C charging limit in a box on the outside wall	Weak	S31 now plausible; moving it into the store proposed (D6)
Sun on an unshielded sensor adds 2 to 3 K	Weak	S32 now names the shield and a field calibration
S34 claimed the scenarios bracket all the inputs; several are not varied	Weak	S34 reworded to the inputs actually varied; a one-at-a-time sensitivity is suggested for the next CAL revision
	Weak	

Finding	Reviewer's verdict	What changed
Option B: 4 K claimed against a 3.2 K estimate; the stack flow runs the other way; 90 % RH does not follow from a moisture balance		B1 now about 3 K; B3 states the direction; B4 now plausible at 55 to 90 %
Option C's cooling cannot be "within known science" while its airflow is open	Weak	C1 now plausible
Bearing on poor soils, under-fired brick, sun on the end walls, mosquito breeding and the floor's ground temperature	Notes	Site soil, brick quality and the other notes are carried into ZBX-RSK-001 and docs/REVIEW.md

Rows the reviewer let stand: S1, S2, S6, S8, S10 to S12, S15, S16, S19, S21, S22, S28, S29, S33, B2 and C3.

Sources

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